ART EQUILIBRIUM UNITY SCRIPTS MANUAL

Script Setup Guide / Air Traffic / Doors / Metro / Camera / Materials

Version 1.0

Quick start

Import the scripts into your Unity project, attach the required script to the correct GameObject, assign all required references in the Inspector, then test the demo scene. If keyboard input does not work, check Unity Player Settings -> Active Input Handling.

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Important
These scripts are intended for demo scenes and simple asset presentation logic. They are not required for using the models, materials, or textures themselves.

1. SCRIPT OVERVIEW

This package includes small helper scripts for presentation scenes and interactive demo content. The scripts can control flying traffic, simple doors, metro trains, platform doors, a free camera, and timed material changes.

File / Script	Purpose	Attach to
CP_AirTrafficLane.cs / AirTrafficLane	Spawns flying vehicle prefabs and moves them between two BoxCollider volumes.	Empty GameObject used as traffic lane controller.
CP_Door.cs / AE_Door	Opens and closes a rotating or sliding door when the Player presses E inside the trigger.	Door object with Collider trigger setup.
CP_MetroTrainController.cs / MetroSplineTrainController	Moves metro cars along a spline path, stops at stations, and opens train doors.	Main train controller object.
CP_MetroStationDoors.cs / CP_MetroStationDoors	Detects the train in a platform trigger zone and opens station doors when the train is dwelling.	Platform door module.
CP_SimpleCameraController.cs / AE_SimpleCameraController	Free camera for scene inspection, screenshots, and demo navigation.	Camera object.
CP_TimedMaterialSwitcher.cs / SimpleMaterialCycler	Cycles through materials over time on a Renderer.	Object with Renderer, screen, light, sign, or effect mesh.

2. GENERAL SETUP

Place the script files anywhere inside the Unity Assets folder. Unity will compile them automatically. Attach each script to the correct GameObject, then assign the missing references in the Inspector.

- Assign all required Transform, Collider, Renderer, Material, and Prefab references manually in the Inspector.
- For trigger-based scripts, make sure the required Collider is set as Trigger and that the Player or Train object can be detected.
- For keyboard input scripts, the included versions use UnityEngine.Input. In Project Settings -> Player -> Active Input Handling, use Input Manager or Both.
- For demo scenes, test the scripts in Play Mode before publishing or recording video.

Safe workflow
Duplicate your demo scene before changing script settings. This makes it easy to test different values without damaging the original presentation scene.

3. AIR TRAFFIC LANE

AirTrafficLane creates a simple flying traffic lane. Vehicles are spawned once at startup, then each vehicle flies from a random point inside Spawn Box to a random point inside Target Box. When it reaches the end, it receives a new route and continues flying.

Basic setup

1. Create an empty GameObject, for example AirTrafficLane_01.
2. Add the AirTrafficLane component.
3. Create or assign two BoxColliders: one for Spawn Box and one for Target Box.
4. Add vehicle prefabs to Vehicle Prefabs.
5. Set Max Active Vehicles, speed, sway, and model rotation settings.
6. Press Play and adjust the boxes in the scene until the traffic direction looks correct.

Setting / Parameter	What it does
Spawn Box	BoxCollider volume used as the random start area for vehicles.
Target Box	BoxCollider volume used as the random destination area.
Vehicle Prefabs	Array of vehicle prefabs. The script randomly chooses one for each spawned vehicle.
Auto Start	If enabled, traffic starts automatically when the scene starts.
Max Active Vehicles	Number of vehicles created and controlled by this lane.
Min Speed / Max Speed	Random speed range for each vehicle.
Random Scale	Enables random scale variation for spawned vehicles.
Min Scale Multiplier / Max Scale Multiplier	Scale range used when Random Scale is enabled.
Side Amplitude	Side-to-side sine movement amount. Set to 0 for perfectly straight flight.
Side Frequency	Speed of the side-to-side sway.
Vertical Amplitude	Up-and-down sine movement amount. Set to 0 to disable vertical sway.
Vertical Frequency	Speed of the vertical sway.
Model Yaw Offset	Extra Y rotation correction if the vehicle model faces the wrong direction.
Force Kinematic Rigidbody	If enabled, all Rigidbody components on spawned vehicles become kinematic and gravity is disabled.
Disable Colliders On Vehicles	Disables vehicle colliders after spawning. Useful for background traffic.
Spawned Parent	Optional parent Transform for all spawned vehicles.

Recommended values

For background flying traffic, start with Max Active Vehicles 10, Min Speed 8, Max Speed 14, Side Amplitude 0.2-0.4, Vertical Amplitude 0.1-0.3, and Disable Colliders On Vehicles enabled.

4. AE_DOOR

AE_Door is a simple interactive door script. When the player enters the trigger area and presses E, the door opens or closes. The script supports both rotating doors and sliding doors.

Basic setup

7. Add the AE_Door script to the door object.
8. Make sure the player object has the tag Player.
9. Make sure the door area has a trigger collider that can detect the player.
10. Choose rotating door or sliding door using Is Sliding Door.
11. Set the open/close UI messages and optional audio clips.

Setting / Parameter	What it does
Smooth	Door movement smoothing speed. Higher value means faster movement.
Door Open Angle	Opening angle for a rotating door.
Is Sliding Door	If enabled, the door moves by position instead of rotating.
Slide Offset	Local offset from the closed position when the sliding door is open.
Open Message / Close Message	Text shown on screen when the player can interact.
Message Font	Optional font used for OnGUI text.
Font Size / Font Color	Visual settings for the interaction message.
Message Position	Screen position of the interaction message. 0.5 / 0.5 is center.
Open Sound / Close Sound	Optional audio clips played when the door changes state.

Input note

This script uses the E key through UnityEngine.Input. If the project uses only the new Input System, set Active Input Handling to Both or replace the input calls with Input System equivalents.

5. METRO SPLINE TRAIN CONTROLLER

MetroSplineTrainController moves one or more train cars along a smooth path generated from waypoint transforms. It supports acceleration, braking, station stops, dwell time, ping-pong travel, model orientation correction, and train door animation.

Basic setup

12. Create waypoint objects in the scene in route order.
13. Create a train controller object and add MetroSplineTrainController.
14. Assign all waypoint transforms to Waypoints.
15. Assign train car root transforms to Cars in order. The first car is the head car.
16. Set Spacing From Previous for each following car.
17. Assign left and right train doors if the train doors should open at stations.

18. Create Station Stop entries and assign stop point transforms.
19. Use Rebuild Spline from the component context menu if you change route points during setup.

Setting / Parameter	What it does
Waypoints	Visible route points in scene order. The train path is created from these transforms.
Ping Pong	If enabled, the train reverses at the ends of the route. If disabled, it loops.
Samples Per Segment	Spline lookup quality. Higher values create smoother path sampling but use more processing.
Cars	Train car list. Each car has a root Transform and optional door arrays.
Spacing From Previous	Distance from the previous car along the spline.
Cruise Speed	Maximum movement speed in Unity units per second.
Acceleration	How quickly the train speeds up.
Braking	How quickly the train slows down before stations or route ends.
Stop Snap Distance	Distance threshold for snapping exactly to a station stop.
Stop Alignment Offset	Moves the effective stop point forward or backward along the route.
Model Forward Axis	Defines which local axis of the train model is considered forward.
Model Rotation Offset	Extra Euler rotation applied after forward-axis correction.
Door Open Speed	Speed used for opening train doors.
Door Close Speed	Speed used for closing train doors.
Stations	List of stop points. Each station has dwell time and left/right door options.
Play On Start	If enabled, the train starts automatically.
Start Normalized Position	Initial position on the route from 0 to 1.
Start Direction	1 moves forward, -1 moves backward.
Start Moving	If enabled, the train begins at cruise speed instead of starting from zero.

Station Stop settings

Setting / Parameter	What it does
Stop Point	Scene Transform that marks the desired station stop position.
Dwell Time	How long the train waits at this station.
Open Left Doors	Opens the left train doors while dwelling at this stop.
Open Right Doors	Opens the right train doors while dwelling at this stop.

Important
The station stop marker is projected onto the spline. Place stop markers close to the path. If the train stops too early or too late, adjust

Stop Alignment Offset.

6. METRO STATION DOORS

CP_MetroStationDoors controls platform doors. The script checks a BoxCollider zone, finds a MetroSplineTrainController inside it, and opens the platform doors only when the train is dwelling and its train doors are open.

Basic setup

20. Add CP_MetroStationDoors to the platform door module.
21. Assign Trigger Zone. This BoxCollider defines the train detection area.
22. Assign all platform door panels to Doors.
23. For each door, set Open Local Offset in local space.
24. Optional: use Required Tag and Detection Mask to limit what can trigger the doors.
25. Test with the metro train controller. Station doors should open while the train is stopped and train doors are open.

Setting / Parameter	What it does
Trigger Zone	BoxCollider volume used to detect the train. The script forces it to be a trigger.
Required Tag	Optional tag filter. Leave empty if no tag filter is needed.
Detection Mask	Layer mask used by the overlap detection.
Doors	Array of platform door panels.
Door	Transform of an individual door panel.
Open Local Offset	Local movement offset from the closed position to the fully open position.
Open Duration	Time in seconds to fully open platform doors.
Close Duration	Time in seconds to fully close platform doors.

How synchronization works

The platform doors do not open only because a train enters the trigger. They open when the detected train is currently dwelling at a station and at least one train door side is open.

7. SIMPLE CAMERA CONTROLLER

AE_SimpleCameraController is a free camera script for demo scenes, screenshots, and asset inspection. It moves with keyboard input and rotates while the right mouse button is held.

Controls

Input	Action	Note
Right Mouse Button	Hold to rotate camera	Mouse movement controls yaw and pitch.
W / A / S / D	Move forward, left, backward, right	Movement is relative to camera rotation.
Q / E	Move down / up	Useful for flying camera movement.
Left Shift	Sprint	Multiplies movement speed.

Escape	Quit play mode or application	Can be disabled with Quit On Escape.

Setting / Parameter	What it does
Move Speed	Base camera movement speed.
Sprint Multiplier	Speed multiplier while Left Shift is held.
Position Lerp Time	Movement smoothing time. Lower value is more responsive.
Mouse Sensitivity Curve	Controls rotation sensitivity based on mouse delta magnitude.
Rotation Lerp Time	Rotation smoothing time. Lower value is more responsive.
Invert Y	Reverses vertical mouse look direction.
Quit On Escape	If enabled, Escape stops Play Mode in the editor or quits the build.

Input note

This script uses the old UnityEngine.Input API. If your project is set to Input System Package only, switch Active Input Handling to Both or use an updated Input System version of the script.

8. SIMPLE MATERIAL CYCLER

SimpleMaterialCycler replaces materials on a Renderer at a fixed time interval. It is useful for animated signs, blinking screens, warning lights, simple advertisement panels, or material-based effects.

Basic setup

26. Add SimpleMaterialCycler to the object that has a Renderer.
27. Assign Target. If the script is on the same object, it can usually find the Renderer automatically.
28. Add two or more materials to Materials.
29. Set Interval in seconds.
30. Choose Slot Index if only one material slot should change, or leave -1 to replace all slots.

Setting / Parameter	What it does
Target	Renderer whose materials will be changed.
Slot Index	-1 replaces all material slots. Any value 0 or higher replaces only that material slot.
Materials	Material list used for cycling. At least two materials are required.
Interval	Time in seconds between material changes.
Change On Start	If enabled, the first material is applied immediately on start.
Use Shared	If enabled, changes sharedMaterials and may affect other objects using the same material reference. If disabled, uses material instances on this Renderer.

Safe workflow

Keep Use Shared disabled for most scene effects. Enable it only when you intentionally want to modify shared material references.

9. TROUBLESHOOTING

Setting / Parameter	What it does
Vehicles do not appear	Check Spawn Box, Target Box, and Vehicle Prefabs. Make sure at least one prefab is assigned.
Vehicles face sideways	Adjust Model Yaw Offset. Try 0, 90, -90, or 180.
Vehicles collide with scene objects	Enable Disable Colliders On Vehicles for background traffic.
Door does not open	Check Player tag, trigger collider, and that the E key is detected.
Metro train does not move	Check that at least two Waypoints are assigned and Cars[0].Root is assigned.
Train does not stop at station	Place Station Stop close to the spline and use Rebuild Spline.
Station doors do not open	Check Trigger Zone size, Detection Mask, Required Tag, and that the train is dwelling with train doors open.
Camera input causes an error	Use Active Input Handling: Both, or update scripts to the new Input System.
Material does not switch	Check Target Renderer, Materials array length, Slot Index, and Interval.

10. SUPPORT

If you have any issues with the scripts, demo scene setup, render pipeline, materials, or shaders, contact us by email:

Support email
art_equilibrium.studio@mail.ru

Please include this information when contacting support:

- Unity version
- Render pipeline: Built-in, URP, or HDRP
- Script name
- Screenshot of the Inspector settings
- Screenshot or video of the issue
- Short description of what happened

Tip
The more exact the description and screenshots are, the faster the issue can be reproduced and fixed.